Design Overview
Simulation Results - Preliminary
Action Plan Moving Forward

Design, Fabrication and Analysis of an Aircraft Wing Structure

AE322 — AE341

Group 2

Department of Aerospace Engineering
Indian Insitute of Space Science and Technology

February 24, 2026

Contents

1. Design Overview
2. Simulation Results - Preliminary
3. Action Plan Moving Forward

Design Overview

Design Overview
Simulation Results - Preliminary
Action Plan Moving Forward

- The selected airfoil is **WE 3.55-9.3**.
- The overall wing structure consists of **2 spars** and **approximately 40 ribs**.
- **Airfoil Properties:**
  - Maximum thickness: 9.3% of chord
  - Maximum camber: 3.55% of chord
- **Wing Dimensions:**
  - Span: 980 mm
  - Chord: 250 mm

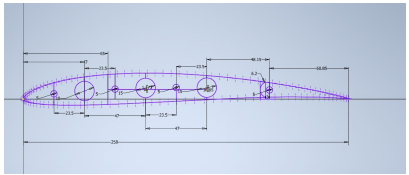

Figure: Airfoil Topology (WE 3.55-9.3)

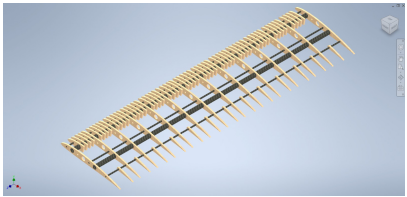

Figure: Rib and Spar Layout

Design Justification

Design Overview
Simulation Results - Preliminary
Action Plan Moving Forward

- **Airfoil Selection – WE 3.55-9.3:** Chosen for its favorable low Reynolds number performance and 9.3% thickness, providing adequate structural depth for spar integration while maintaining efficient lift-to-drag characteristics and stable stall behavior.

- **Structural Configuration (2 Spars, ∼40 Ribs):** Two spars effectively resist bending and torsion, along with Stringers provided while closely spaced ribs (∼40 over 980 mm span) ensure airfoil shape fidelity, and provide uniform load transfer along the span.

Simulation Results - Preliminary

Design Overview

Simulation Results - Preliminary

Action Plan Moving Forward

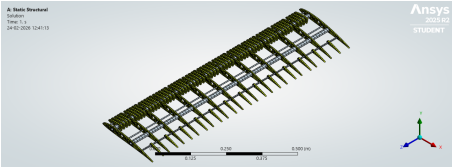

Figure: Finite Element Mesh of the Wing Model

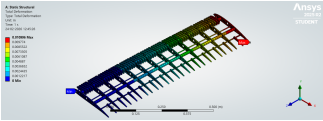

Figure: Total Deformation under 10 N Load

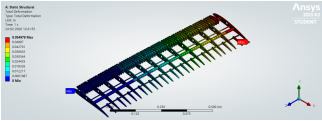

Figure: Total Deformation under 50 N Load

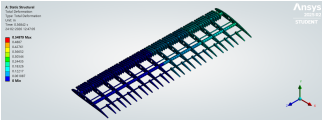

Figure: Total Deformation under 500 N Load

Action Plan Moving Forward

Design Overview
Simulation Results - Preliminary
Action Plan Moving Forward

- **Manufacturing**
  - Rib material: Balsa wood
  - Spar material: Aluminium rods
  - Outer skin (if required): Aluminium sheet
- **Simulations**
  - Vibration Analysis
  - Torsional Analysis
- **Testing**
  - Vibration testing
  - Bending load testing
  - Torsion testing